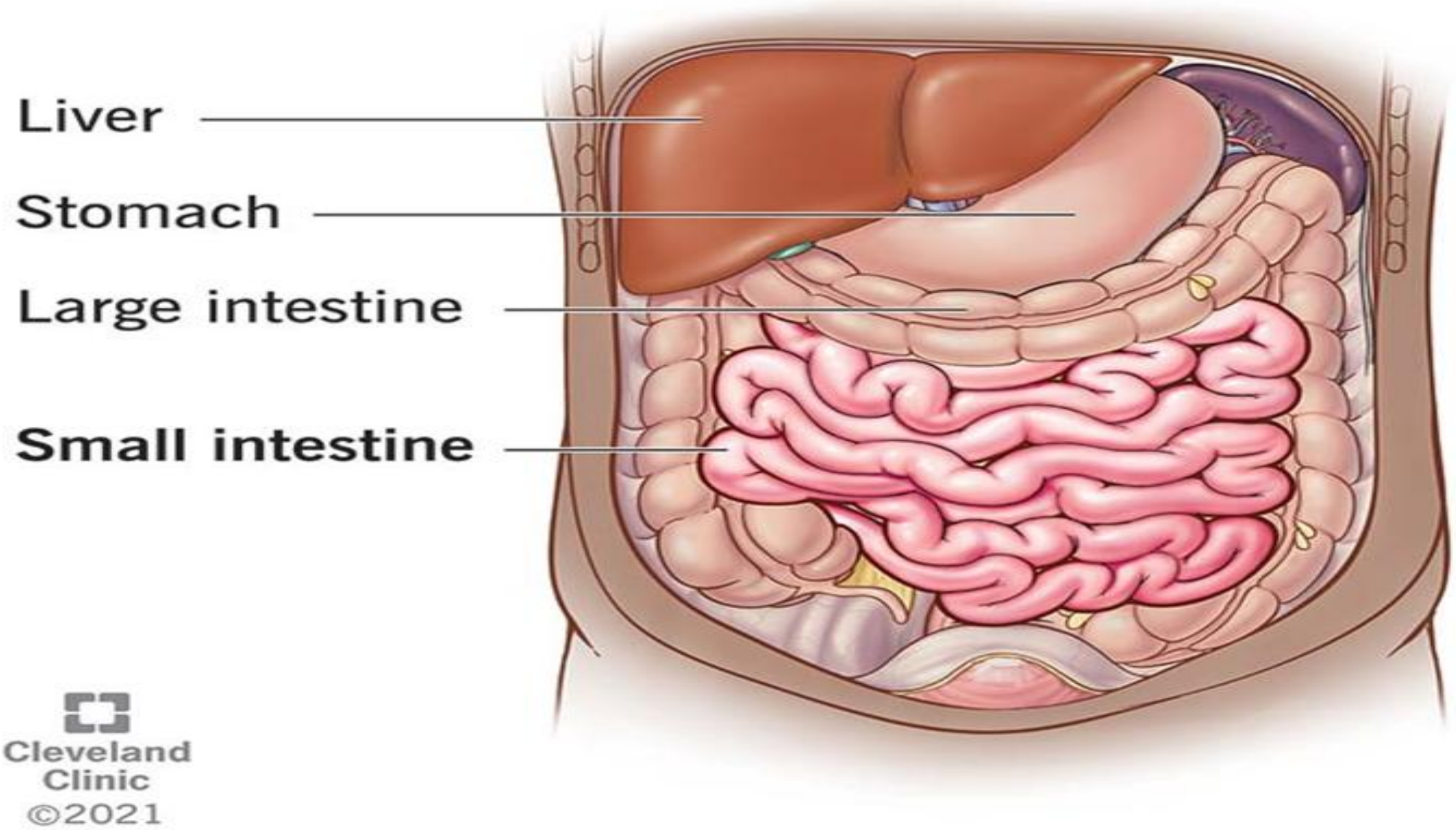
Small Intestine

Medical physiology

ILOs

1. Recognize the parts of the small intestine
2. Describe types of intestinal motility and its control.
3. Describe the function of ileocecal valve.
4. Explain regulation of intestinal motility
5. Describe functions of intestinal secretion

Small intestine



Small Intestine

It is the part of gastrointestinal (GI) tract, extending between the pyloric sphincter of stomach and ileocecal valve, which opens into large intestine. Its length is about 6 meter.

Small intestine consists of three portions:

1. Proximal part known as duodenum
2. Middle part known as jejunum
3. Distal part known as ileum.

Important function of small intestine

is absorption.

Maximum absorption of digested food products takes place in small intestine.

Motility (movement) of the small intestine:

Intestinal motility 1. Mix intestinal contents

2. Propel it towards large intestine

- The intestinal slow waves (**Basic Electrical Rhythm**) (BER):
- These are slow waves of **partial** depolarization that occur at a rate:

❶ 12 waves/minute at duodenum

❷ 8 waves/minute at ileum

- They start at "Pacemaker of small intestine" around the ampulla of Vater where they proceed forwards
- When the gut is distended mechanically or under the effect of vagal impulses, the slow waves produce action potentials that result in mechanical response

Types of movements of small intestine:

❶ Segmentation (mixing) contractions *(Most frequent type)*

- **Description**:
  - Several constrictions divide a loop of intestine into equal segments
  - A moment later, new constrictions occur in the middle of each segment
  - The frequency of the segment contractions is:
    - ❶ 12 contractions/min in duodenum
    - ❷ 8 - 9 contractions/min in terminal ileum

- **Control**: 1. Myogenic: they are controlled by the slow waves (BER) in the smooth muscles of the intestine
  2. Enteric nervous system

- **Function**: 1- Digestion: They mix the chyme with digestive juices
  2- Absorption: They allow food to come in contact with the mucosa facilitating absorption

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❷ Peristaltic movement:

- Mainly propulsive
- **Description**: Intestinal distension leads to: ❶ Contraction behind
  ❷ Relaxation in front
- They propagate for few cms then die out
- They move the chyme at a rate of 1cm/min
- They are faster in the proximal intestine & slower in the terminal intestine

- **Normally:** Passage of chyme from **pylorus** to **ileocecal valve** requires **3-5 hours**
- **Control** 1. Peristalsis depends on the integrated activity of the enteric nervous system
  2. The initiating stretch of the wall
  3. The wave of contraction behind the stimulus is produced by:
     ❶ Acetyl choline
     ❷ Substance P
  4. The relaxation in front is produced by: VIP

  Function of peristalsis: Propagation of food

Intestinal secretion (Succus entericus).

PROPERTIE OF Succus entericus

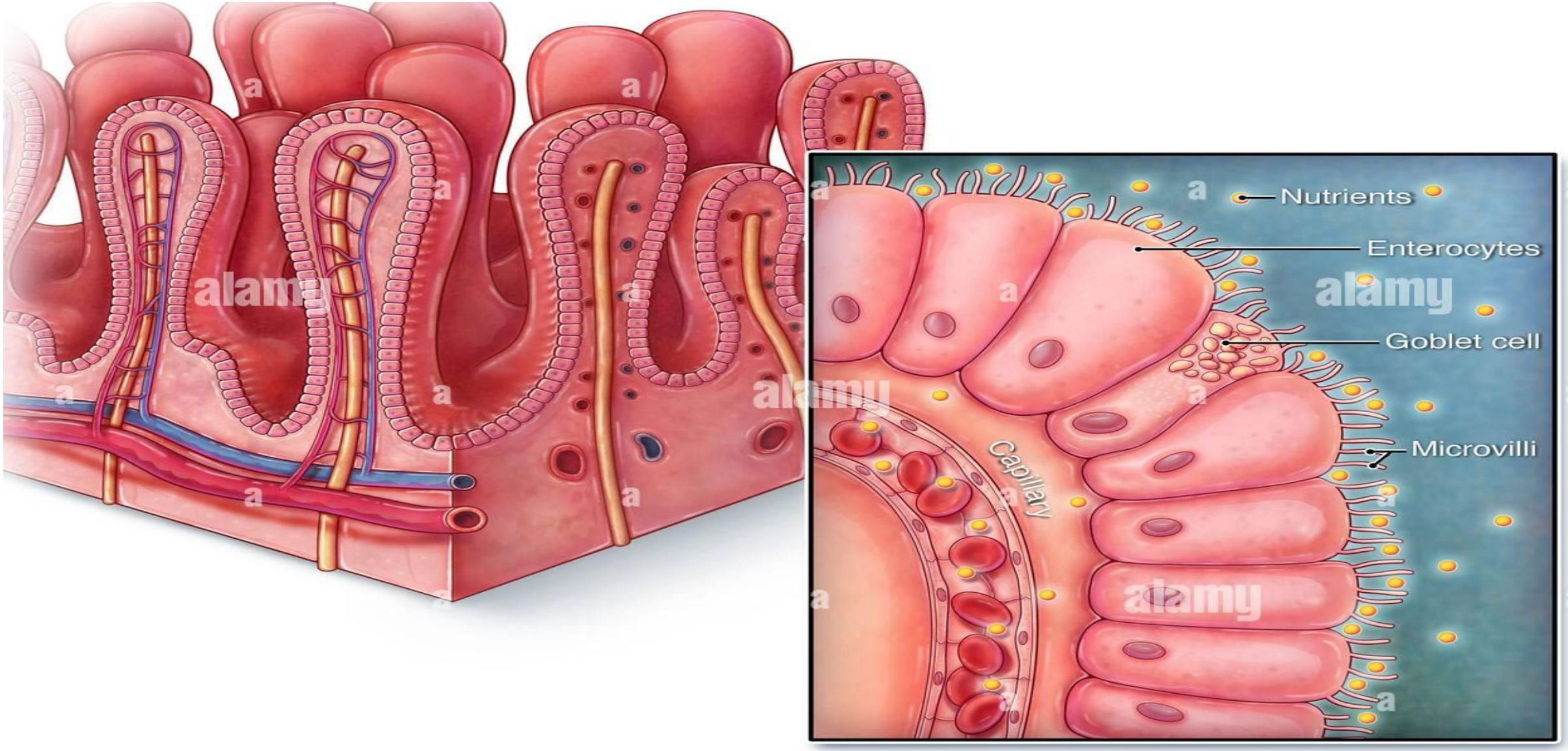
Volume : 1800 mL/day

Reaction : Alkaline

pH: ٨,٣

COMPOSITION OF Succus entericus

Succus entericus contains water (99.5%) and solids(0.5%) (Solids include organic and inorganic substances . Bicarbonate concentration is slightly high in succus entericus.



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FUNCTIONS OF SUCCUS ENTERICUS

1. DIGESTIVE FUNCTION

Enzymes of succus entericus act on the partially digested food and convert them into final digestive products.

Enzymes are produced by enterocytes of the villi

a) Proteolytic enzymes present in succus entericus are the peptidases, that convert peptides into amino acids

b) Amylolytic enzymes of succus entericus

Lactase, sucrase and maltase convert the disaccharides (lactose, sucrose and maltose) into two molecules of monosaccharides

a) Lipolytic Enzyme

Intestinal lipase acts on triglycerides and converts them into fatty acids.

2. PROTECTIVE FUNCTION

Mucus present in the succus entericus protects the intestinal wall from the acid chyme, which enters the intestine from stomach; The antimicrobial peptides secreted by intestinal glands .These peptides are called natural peptide antibiotics because of their role in killing the phagocytosed bacteria

3. ACTIVATOR FUNCTION

Enterokinase present in intestinal juice activates trypsinogen into trypsin. Trypsin, in turn activates other enzymes .

4. HEMOPOIETIC FUNCTION

Intrinsic factor of Castle present in the intestine plays an important role in erythropoiesis It is necessary for the absorption of vitamin B12.

REGULATION OF SECRETION OF SUCCUS ENTERICUS

Secretion of succus entericus is regulated by both nervous and hormonal mechanisms.

1) NERVOUS REGULATION

Stimulation of parasympathetic nerves causes vasodilatation and increases the secretion of succus entericus.

Stimulation of sympathetic nerves causes vasoconstriction and decreases the secretion of succus entericus.

However, the local nervous reflexes play an important role in increasing the secretion of intestinal juice. When chyme enters the small intestine, the mucosa is stimulated by tactile stimuli or irritation. It causes the development of local nervous reflexes, which stimulate the glands of intestine.

2) HORMONAL REGULATION

When chyme enters the small intestine, intestinal mucosa secretes enterocrinin, secretin and cholecystokinin ،which promote the secretion of succus entericus by stimulating the intestinal glands.

Thank You